Energy Life Cycle in Extraterrestrial Construction Projects: Is Building Underground an Option?

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ABSTRACT

Making life multi-planetary requires substantial knowledge of building structures in extraterrestrial bodies, where natural resources are scarce, human resources are expensive-to-utilize, and energy sources are vastly different from the Earth. On Earth, fossil fuels have been the main source of energy for construction projects. In other extraterrestrial bodies, including Mars and Moon, easy-to-access fossil fuels are non-existent. Solar energy is, and probably will be, the primary source of energy accessible to extraterrestrial construction projects. This calls for ultra-efficient and sustainable design of structures and buildings, smart and resource-efficient planning of projects, and use of energy-efficient and smart construction techniques and machinery. This paper presents two energy cycle flowcharts for construction projects on Earth and beyond, which show the vastly different energy cycles in these environments. It further presents arguments about the expensive and unrealistic nature of building underground, as one of the key ideas for building beyond Earth.

INTRODUCTION

Humans have relied on safe refuges to protect themselves and their possessions from the surrounding environments they live in ever since man was known to exist (Horning 2009). The types of structures vary greatly depending on the culture, region, local climate, availability and types of available building materials and individual needs (Leslie 2010; Wallis 1997). In near future, humankind will move beyond earth to extraterrestrial bodies in the solar system. As our ancestors millions of years ago, again, we will require protection from the harsh extraterrestrial environments found on these solar system planetary and moon bodies. Future shelters will have to be designed to adequately protect occupants from extreme environmental conditions; including no atmospheres, high or low atmospheric pressures, meteorite or asteroid impacts, lack of protective magnetic fields, extreme temperature changes and harmful radiation emanating from our Sun and interstellar space. These shelters will also need to provide livable and usable spaces that enable humans to live and work safely (Kennedy 2002). Colonization of the Moon or Mars would be considered an excellent starting point for humans to learn how to adapt to living

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outside of generally comfortable Earth atmosphere. For a successful inhabitation transition there may be a need to use different phases and types of construction models that evolve progressively based on expected needs and lengths of time. Some scientists have argued that this could be done over a three-phase development program. The initial phase is for short term missions. The second phase where structures would be designed for slightly longer lengths of stays. Finally, the third phase where structures would be built for long term and permanent living (Jablonski and Ogden 2008).

In order to permanently inhabit a planet, humans need to build permanent structures that last for decades. While it is argued that it is possible to manufacture permanent, and possibly modular, structures on Earth and ship to and assemble in other planets, it may not be economically feasible to do so for large scale colonization. The Earth's gravity makes transporting heavy loads into space an expensive and complicated exercise. Sophisticated space vehicles are required to launch heavy equipment into space by using chemical rocket propulsion (Mueller et al. 2016). Commonly, rockets use liquid hydrogen as the fuel source for power (Peschka 2012). Liquid hydrogen for rocket fuel is mainly produced from natural gas, a fossil fuel (Spath and Mann 2000). Therefore, a significant amount of fossil fuel must be used for transportation of materials which makes this option very expensive and, most certainly, infeasible for large scale colonization (Bonanno et al. 2012). Aside from materials, multiplanetary life needs a reliable source of energy. Without energy, there will be no life anywhere in the solar system. One easy way of solving multiplanetary energy dependency issue is to transport fossil fuels from Earth. However, like transportation of materials, transportation of energy will be a very expensive option and, most certainly, infeasible for large scale colonization. It makes sense to utilize solar energy as the most common and readily available form of energy available in our solar system to solve the multiplanetary energy dependency issue.

Utilizing solar energy as the main source of energy in multiplanetary life has some drawbacks, including low temperature, low intensity, indirect sunlight and long nights on some bodies of solar system such as Mars (Landis 2000). But it also comes with some advantages over other renewable and non-renewable energy sources including its constant abundance on most planetary and moon surfaces, easy accessibility, predictability and price; solar energy comes with only a down payment and without considerable recurring charges. Using solar energy, as the main source of energy, for purposes such as building houses, refuges and structures is not an easy task. It asks for (1) ultra-extreme efficient processes, (2) bold ideas on harvesting, storage and transportation of solar energy and (3) audacious energy-conservative building planning, which is the topic of this paper. This paper presents two energy cycle flowcharts for construction projects on Earth and beyond, which show the vastly different energy cycles in these environments. It further presents (1) a discussion on availability of solar energy on Mars, and (2) estimate the amount of solar energy needed for only removal of Martian soil needed for digging for building a small structure on Mars. This simple case study indicates the considerable energy consumption needed for removing Martian soil; consequently arguing that building any type of underground structure that needs extensive soil removal might not be a feasible building option.

IMPORTANCE OF ENERGY IN CONSTRUTCION PROJECTS

Energy in building construction processes on Earth. The process of constructing buildings, structures, infrastructures and other civil structures that support life on Earth is called

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Construction Management (CM). CM overseas the whole built process from start to finish. One of the main tasks of construction managers is to manage the elements of iron triangle in projects which include *time*, *cost* and *scope*. Construction managers are responsible for completing projects on time and on budget and also by keeping all stakeholders happy during the lifetime of projects. But in CM, there is little, if any, discussion about the importance, cost and availability of energy for projects. The widespread availability of fossil fuels and electricity, unusually cheap cost of energy and the vast network of energy distribution on Earth made the issue of energy absolute in planning of construction projects. Unless a project is built on a remote area, which due to prevalent access to the internet and availability of great infrastructures on Earth rarely happens, energy is not a top priority in construction projects. Relatively easy access to fossil fuels, which almost all construction equipment and machinery rely on, and easy access to energy enabled men to build structures everywhere on Earth, with different types of ideas. Vast highway networks, massive underground subway systems and strangely tall skyscrapers are only some of the building ideas that men had and successfully executed. With constant access to cheap energy, men even built the most expensive structure on Earth and send it to space as *International Space Station*. However, without access to cheap and reliable energy, all these would not be possible.

Unfortunately the easy access to energy made the issue of planning for energy an absolutely irrelevant issue in the process of managing construction processes. For examples, on earth making a decision on building a structure above ground versus underground is not restrained by the amount of energy needed but by the needs of the client. Unfortunately, while in the last 50 years, life on Earth in general and CM in particular were free of any discussion regarding energy scarcity, the future of built environment will definitely be different; with more regards to energy conservation. The whole CM process, with regard to energy, on Earth is illustrated in Figure 1. As Figure 1 demonstrates, energy is an integrated part of building processes; however, in reality, and *sadly*, it is considered one of the least-important subjects.

Energy in building construction processes on Space. It is argued that multiplanetary colonization could happen in a three-phase development program: (1) temporary living in structures for short periods during short term missions, (2) living inside structure that are designed for slightly longer lengths of stays and (3) permanent colonization in structures built for long term, or permanent, stays (Jablonski and Ogden 2008). Many studies have been conducted on the design of prefabricated and inflatable structures required for the first phases of colonization. Criswell and Carlson (2004) proposed building an inflatable three-level spherical dome with reinforced edges and extendable reinforcing elements such as beams and rings that had diameters of 9 meters. Also, an inflatable structure called Sasakawa International Center for Space Architecture (SICA) was designed for more comfort and space while being versatile and adapted for long-duration missions. SICA is a spacious three-level structure wrapped around a small core which provides circulation (Zatakia 2006). Using inflatable structures inside natural terrain features such as lava tubes and caves would be a logical choice for the second phase, because of the cost-effectiveness and high protection (Billings et al. 2000). Inflatable structures are compact, light and flexible, making them economical to transport and can be made to any specification including the levels of protection needed, including the use of different wall layers and linings. Inflatables are versatile in the way that other modules can be fitted onto existing structures (creating multi-level structures) and can be integrated into more permanent, stable structures of the proposed future phases of colonization (Zatakia 2006).

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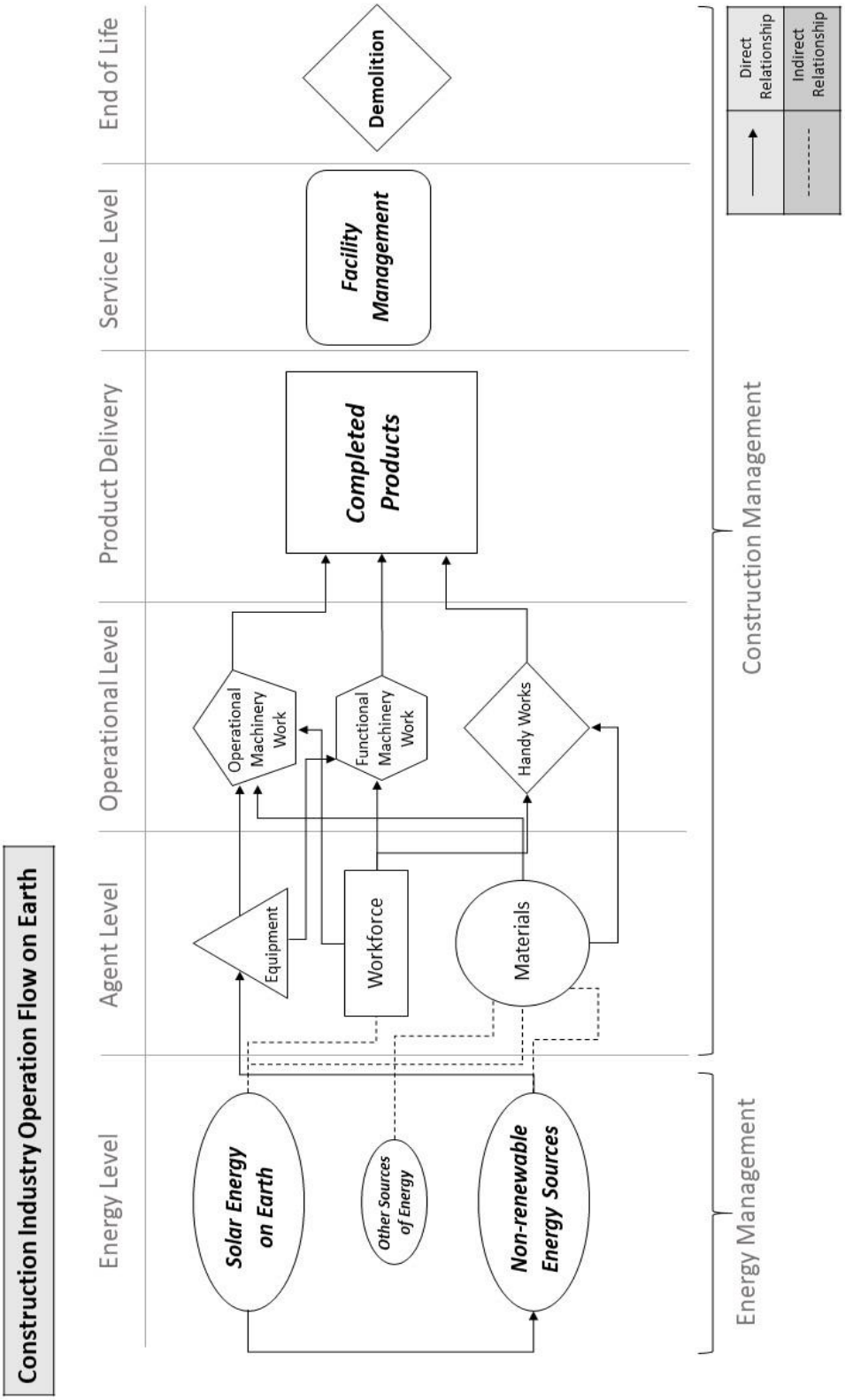

Figure 1. Energy cycle flowchart in construction projects on Earth.

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Finally, the third phase, where more permanent structures could be built on either the surface, inside caves, lava tubes or underground have been studied by many scholars. The advantages of using the natural caverns including lava tube, caves and so forth are the protection they can offer to the structures from dangerous solar radiation, asteroid and micro-meteorite impacts and the extreme variances in temperatures experienced on the exposed planet or moon surface (Rostami et al. 2018). In the early 1960s, lava tubes were recognized as protection shelters and locations for habitation (Green 1963). In 2010, they gained interest again when natural lunar cave openings and skylights were discovered (Haruyama et al. 2010). Billings et al. (2000) have investigated various lava tube construction options at three development stages. The first level is accessing natural caverns. It is possible to enter the natural openings easily, but more likely the openings are exposed to regolith's which need to be excavated by small earthmoving vehicles including tractors or dragline muckers. Then for a maturing settlement, many elements including steel, iron, glass, and bricks need to be produced in situ for construction. However, for scarce materials, we would be still dependent on the earth. The use of regolith and natural caverns would help to minimize the need for mechanical excavation of the planet or moon's surface by heavy equipment, which would be expensive to transport to space.

Eventually, the need for larger and heavier material moving equipment will be required because of the efficiency of such equipment in specific applications. A study into modifying large pieces of machinery, such as the Tunnel Boring Machine (TBM), for the use specifically on the lunar surface was done (Rostami et al. 2018). Rostami et al. (2018) proposed modifying the existing technology of TBM's to creating a machine that can be used on the lunar surface, a Lunar Tunnel Boring Machine (LTBM). Using LTBM's can help expedite the creation of useable spaces for inflatable structures or domes by quickly excavating and clearing lava tubes of any debris and creating level surfaces suitable for the structures intended. While using TBM might seem a great idea, running TBM requires a lot of energy; in terms of fossil fuels transported from Earth or mined in Mars or use of renewable energy sources on Mars. The process of transporting construction equipment, workers and building materials, and operating construction equipment consumes great energy (Yan et al. 2010). This paper studies the energy needed for digging a small structure on Mars in order to show the infeasibility of using TBM or any building technique that requires soil excavation.

ENERGY NEEDED FOR BUILDING A SMALL STRUCTURE ON MARS

Energy is defined in this context as the ability to do work. Humans rarely think about energy in their daily lives. Human being considers energy as a cheap source that is abundant available on Earth. Fossil fuels is the main source of energy that fuels our economies. It is chemical bound in various hydrocarbon forms for humanity to extract. Alternative energy sources are harvested from renewable sources. In construction on Earth, the primarily means of energy is fossil fuels. Currently there are no know fossil fuel deposits on Mars or Moon. Even if there would be carbohydrates fossil fuels, there would be no practical way to burn it due to lack of oxygen presence, which itself is a precious resource in outer space. While this might change, currently we have to assume that all energy we need for building on other planets must be transported there or harvested onsite. Energy scarcity would be a real threat to multiplanetary life. One easy way to fix it is to transport energy from Earth. Due to the delivery process from earth surface, energy on Mars is bizarrely more expensive than on Earth. For example, current estimates of delivery cost from Earth to Mars are in the Range of 15-45 thousand dollars per kilogram (KG);

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depending on using Falcon Heavy or Atlas V 541 (SpaceX, 2012). Thus, delivery of a single liter of conventional fossil fuel to Mars costs around $11,233 (0.7489 kg per liter * $15,000 = $11,233). A gallon of fuel may cost between $45,000 and $170,000 on Mars, which makes one kilowatt hour (kWh) to costs in the range $1,158 on Mars. In comparison with the average cost of $0.12 per kWh in the United States, transporting energy from Earth to Mars would be costly. With regard to this massive cost, lack of oxygen as a competitive resource in outer space and long delivery time, due to distance between Earth and Mars, transporting fossil fuels from Earth to Mars does not portray to be a feasible energy supply option. It is argued that it is possible to use (1) nuclear batteries and (2) nuclear reactor, shipped from Earth, as energy sources on Mars. This paper disregard use of these two options due to immaturity in research regarding the feasibility and sustainability of their use as reliable energy sources on Mars. It appears that feasible energy sources on Mars are (1) solar energy through photovoltaic (PV), (2) solar thermal, (3) wind (Haslach 1989) and (4) geothermal (Schulzemakuch et al. 2007).

Mars has a very thin atmosphere, almost 24 hour day cycles and is fairly close to the sun, which make solar energy a viable option; as many robotic missions to mars have already proven. Rovers powered with PV cells lasted well over their specific mission target durations. The best know solar powered object on Mars is probably the Rover Opportunity, which exceeded its design of 90 sols (1 Sol is a Martian solar day,) by almost a factor of 57 and stopped working after 5111 sols (Mars Exploration Rovers 2019).

In order to show the different nature of energy in extraterrestrial construction projects versus projects on Earth, Figure 2 is presented. Figure 2 shows the flowchart of energy cycle in extraterrestrial construction projects; considering the integration of robotic agents and automated system into construction processes.

CASE STUDY

In order to find the exact amount of energy needed for building purposes on Mars, a case study is presented that estimates the energy needed for building a small underground room, which needs digging and removal of soil for a total volume of 600 cubic meters (m3). This paper is not concerned with the details, engineering, functionality or delivery of materials. But only focuses on estimating and comparison of energy needed for building this room underground. In an excavation process, depending on the stability of the soil ramps and slops are needed for soil removal purposes. The layout of the proposed layout of the excavation is shows in Figure 3. As shown in Figure 3, it is assumed that two ramps are used for the purposes of removing soil by heavy construction equipment. These two ramps need to be refilled in a process that is called backfill in construction vocabulary.

$$Total\ soil\ removed = \\ room\ size\ volume + ramp(s) volume + backfill\ of\ ramp(s)\ volume \quad (1)$$

The total volume of soil that needs to be removed is equal to $1200\ m^3$. , as shown in Formula 2.

$$Total\ soil\ removed = 600 + (2 * 300) + (2 * 300) = 1800\ m3 \quad (2)$$

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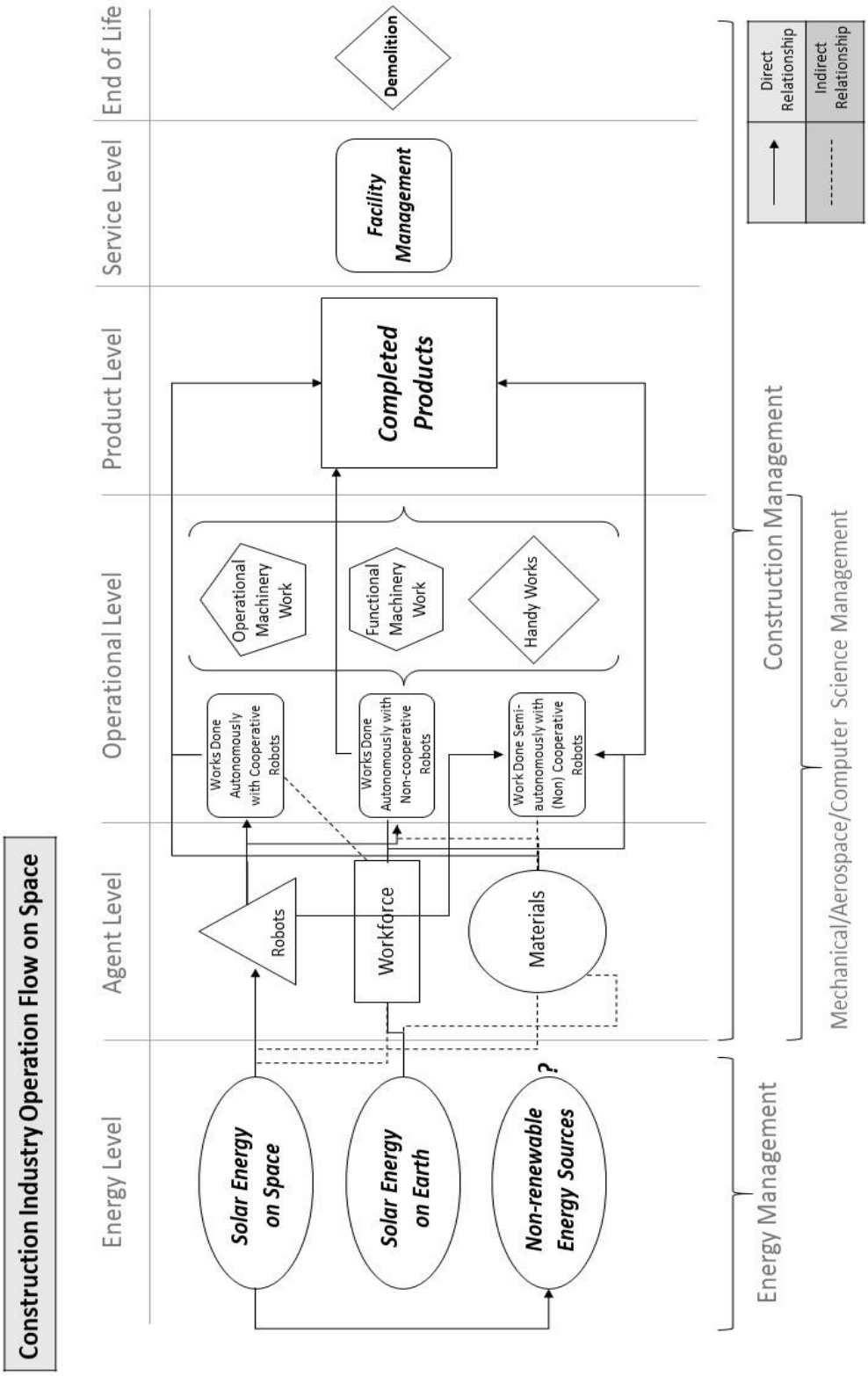

Figure 2. Energy cycle flowchart in extraterrestrial construction projects.

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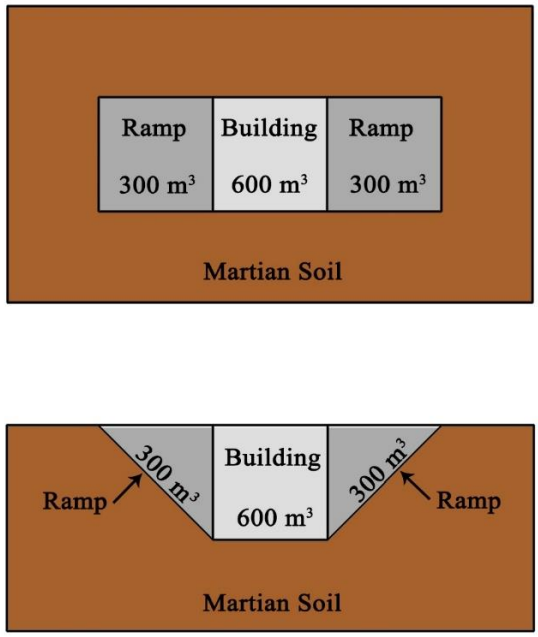

Figure 3. Plan and elevation view of underground room.

$Total\ soil\ removed$ =The energy needed ***only*** for excavation of one cubic meter of soil on ***Earth*** and loading the excavated soil into a dump truck varies between 12.09 *MJ/m3* or 3.36 kWh for Silty Sand and 89 *MJ/m*^{3} or 24.72 kWh for Rock. The variation happens due to use of different types of excavation equipment and different types of soils (Devi and Palaniappan 2017). In this case study, it is assumed that 12.09 *MJ/m3* of energy is needed for removing soil. For excavating the total volume of 1800 m^2 soil, ***21762 MJ*** or 6045 kWh (1800*12.09). This energy would be equal to 181 gallons of gasoline; considering one US gallon is equal to 33.41 kWh.

Solar energy available on Mars. Maximum Solar Radiance on Mars is assumed to be around 590 W/m^2 (Appelbaum and Flood 1990). The yearly mean of solar radiance is close to 200W/m^2. The length of one Martian day is 24.62 hours and duration of daylight is around 12.13 hours or 728 minutes. Even with trackable solar panels we are not expecting more than 90% of useful daylight. Thus a 90% factor is inserted in Formula 3. The efficiency of solar panel is assumed to be 30%; as 30% efficiency is the best that some of solar panels with cutting edge technology can provide. Formula shows the estimated amount of solar energy that can be collected by solar panels on Mars.

$$12.13\ \frac{h}{d}\ x\ 90\%\ x\ 200\frac{W}{m^2 x}30\% = 655.20\ Wh/m^2/d \tag{3}$$

This assumption is close to current NASA InSight lander solar data. According to NASA publication, InSight yielded roughly *604Wh/m²/d* on its first day; with use of solar panels that have the efficiency of *29.5%* (Weaver 2018).

It is assumed that a square solar field with sides of *20 m* is available; yielding *400 m²* of solar panels. This field on average collects solar energy equals to *264 kWh per Martian day* (400*0.6552). In this case study, excavation process of the proposed 600 m^3 underground room

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needs ***6045 kWh*** energy. By having a 400 *m²* solar field available, it takes around ***29 Martian Days*** to produce this excavation energy. It is possible to harvest the full ***6045 kWh*** energy that is needed for excavation in one Martian day if a solar field of ***11424 m²***, or ***2.822 acres*** of solar panels, is available.

DISCUSSION AND CONCLUSION

Using sun as an energy source has many advantages over other renewable solutions, which include its abundance on most planetary and moon surfaces. Having a renewable source like solar energy means having the ability to provide continuous energy supply for all needs, and the construction machinery and processes. Setting up a solar-powered energy system would be more cost-effective than other sources of energy. Photovoltaics are a great option since they are inexpensive, light and easily transportable (Badescu 1998). While there are other types of renewable energy, such as nuclear and wind, each of these have disadvantages that make them economically inviable or unsafe source of energy. For instance, although nuclear energy could produce thermal energy and electrical energy at a continuous rate during day and night, it produces toxic waste products. Moreover, developing nuclear energy is so expensive and requires a long period of time. Another energy source option that may be considered is wind power. However, this has limitations too. There are environmental factors to take into consideration when using wind power as a source of energy. An example would be the lack or shortage of wind on some of our planetary surfaces. Mars has a low atmospheric density and low wind speed, which makes wind an inappropriate power source. Also, the large size of a wind turbine system required inhibits this as a viable option beyond our Earth's atmosphere (Martín-Torres et al. 2015).

A brief case study presented to show the amount of solar energy needed for soil excavation of a small-size underground building. Based on the calculations provided in this paper, excavation for a 600 m^2 building on Mars asks for almost 29 days of solar energy from a 400 m^2 PV field on Mars. This just shows the extravagant amount of solar energy that is needed for only excavation and backfill of a small underground structure. The main conclusion of this paper is impracticality of unusual excavation on Mars. While using excavation equipment such as TBM might seem like a practical way to build structures underground, it is usually forgotten that the amount of energy that large-scale construction equipment such as TBM needs is massive. Without fossil fuels use of such large-scale equipment is not an option on Mars. Developing multiplanetary life depends on building structures in massive scale on other planets. Developing new structures on other planets is restricted to the amount of harvested, or transported, energy on site. Considering lack of fossil fuels, it appears there is only one way to develop cities beyond Earth: use a combination of (1) ultra-efficient and sustainable design of structures and buildings, (2) smart and resource-efficient planning of projects and (3) use of energy-efficient and smart construction techniques and machinery.

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